6 Ans: B

Before coupling breaks,

$F_{net} = ma$

$1.2 \times 10^5 = 2 (1 \times 10^5)\ a$

$a = 0.60\ m\ s^{-2}$

After 20 s, both front and back carriages will be at $v = 0.60\ (20) = 12\ m\ s^{-1}$

After coupling breaks,

$F_{net} = ma$

$1.2 \times 10^5 = (1 \times 10^5)\ a$

$a = 1.2\ m\ s^{-2}$

Using $s = ut + \frac{1}{2} a t^2$, calculate the distance travelled after coupling breaks for the 2 carriages separately

$s_{back} = 12 (20) + \frac{1}{2} (0) 20^2 = 240$

$s_{front} = 12 (20) + \frac{1}{2} (1.2) 20^2 = 480$

Distance apart = 480 - 240 = 240 m